

Fundamental of AI

(BE02000041)

Unit – 2

Knowledge Representation - Introduction to Expert Systems

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Outline

- What are Expert Systems?
- Key Components of Expert Systems
- How Expert Systems Work
- Types of Expert Systems
- Applications of Expert Systems
- Advantages & Challenges
- Future of Expert Systems

What Are Expert Systems?

Definition:

- An expert system is a computer program that mimics the decision-making ability of a human expert.
- It uses knowledge and inference rules to solve complex problems in specialized domains.

Purpose:

- To provide solutions or advice based on expert knowledge in fields like medicine, engineering, finance, etc.

Example:

- A medical diagnostic system that suggests possible diseases based on symptoms.

Key Components of Expert Systems

Knowledge Base:

- Contains facts and rules about the domain (e.g., medical knowledge, legal principles).

Inference Engine:

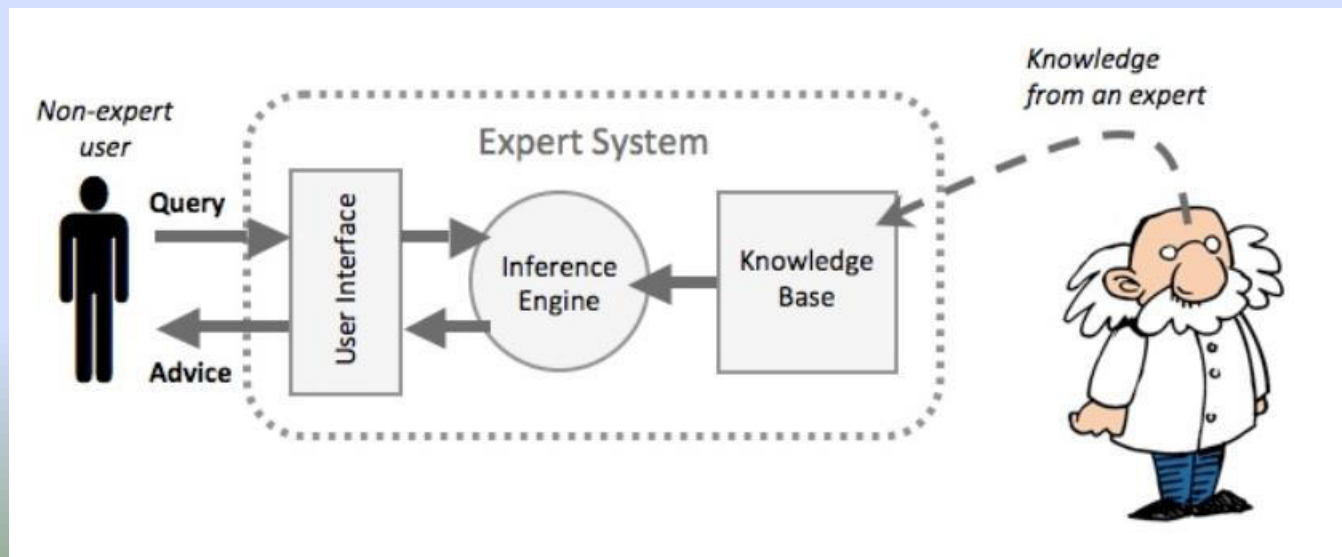
- The "brain" of the system, it applies logical rules to the knowledge base to draw conclusions.

User Interface:

- Allows users to interact with the system (input questions, get answers).

Explanation System:

- Explains how conclusions were reached based on the rules and facts in the knowledge base.



How Expert Systems Work

Input:

- The user provides input, often in the form of symptoms, observations, or problems.

Inference:

- The system applies its knowledge and inference rules to process the input.

Output:

- The system produces conclusions, solutions, or recommendations.

Explanation:

- The system may offer a justification for its conclusions to the user.

Types of Expert Systems

Rule-based Expert Systems:

- Uses "if-then" rules to derive conclusions.
- Example: MYCIN, a medical diagnosis system.

Frame-based Expert Systems:

- Uses structured data (frames) to represent information.
- Example: Systems for design and troubleshooting.

Model-based Expert Systems:

- Uses mathematical models to simulate processes or systems.
- Example: Expert systems for control in industrial processes.

Hybrid Expert Systems:

- Combines different techniques (e.g., rule-based, frame-based).

Applications of Expert Systems

Healthcare:

- Diagnosis support systems (e.g., MYCIN, INTERNIST).

Finance:

- Investment advisory systems, credit scoring systems.

Engineering:

- Troubleshooting and maintenance systems for machinery.

Law:

- Legal expert systems that offer advice based on laws and regulations.

Customer Support:

- Chatbots and virtual assistants that answer queries.

Advantages of Expert Systems

Availability:

- They provide 24/7 service and decision-making capability.

Consistency:

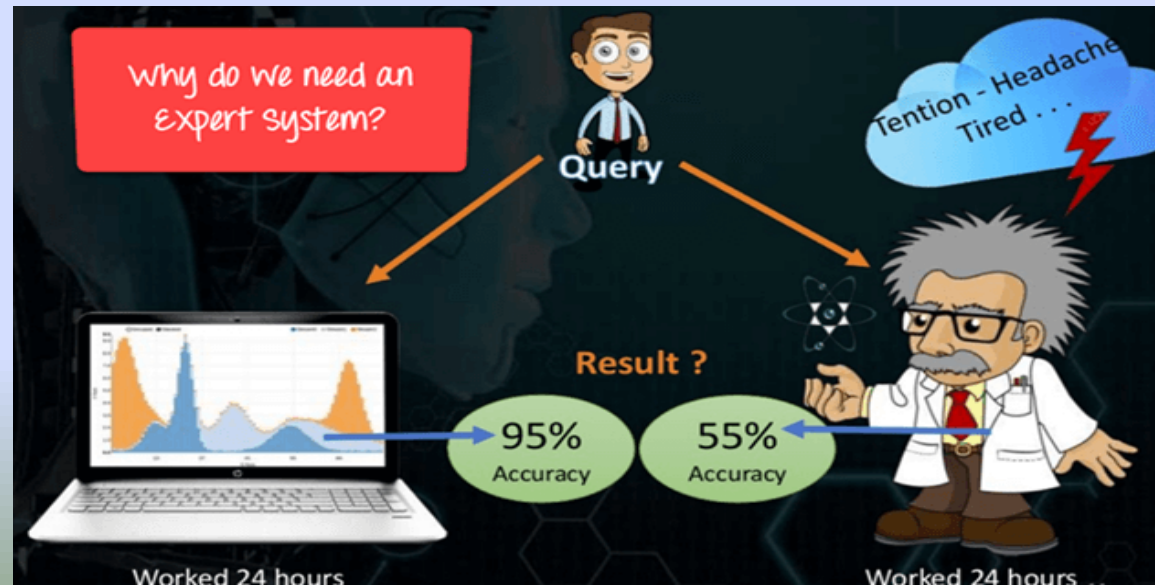
- Expert systems always apply the same rules, ensuring consistent decisions.

Knowledge Preservation:

- Experts' knowledge can be stored, protecting it from being lost.

Efficiency:

- Can solve complex problems quickly, reducing the need for human intervention.



Challenges of Expert Systems

Limited to Specific Domains:

- Expert systems are designed for a narrow field and cannot generalize to other domains.

Knowledge Acquisition:

- Gathering expert knowledge can be time-consuming and difficult.

Maintenance:

- Expert systems require regular updates as knowledge evolves.

Interpretability:

- Some systems might struggle to explain why a certain conclusion was reached.

Future of Expert Systems

Integration with Machine Learning:

- Future expert systems will leverage AI techniques like machine learning to enhance their decision-making abilities.

Natural Language Processing (NLP):

- Improving communication between users and expert systems in natural language.

More Autonomous Systems:

- Expert systems will become more autonomous, able to make more sophisticated decisions with minimal human intervention.

AI-powered Virtual Experts:

- Potential for AI to create expert systems in areas where human experts are rare or hard to access.

Conclusion

Summary:

- Expert systems play a critical role in AI by emulating human expertise in decision-making. They help solve complex problems across various domains.

Key Takeaway:

- While expert systems have some limitations, they remain invaluable tools in industries requiring expert-level knowledge and decision support.